

End Semester Examination Dec. 2024

Name of the course: B. Tech

Semester: I

Name of the Paper: Basic Electronics Engineering

Paper Code: **TEC 101**

Time: 3Hours

Maximum Marks: 100

Note:

- (i) All questions are compulsory
- (ii) Answer any two **questions** among a, b, & c **in each main question**.
- (iii) Total marks in each main question are twenty
- (iv) Each question carries 10 marks

Q1	(20 marks)	
(a)	Solve the following function with the help of K-Map and realize the simplified equation using NOR gate only: $F(A, B, C, D) = \sum m(1,3,4,6,8,9,11,13,15) + \sum d(0,2,14)$	CO1
(b)	State and prove De Morgan's theorem. Explain min-terms and max-terms.	
(c)	(i) $(C8B63.0B)_{16} = (?)_7$ (ii) The solution of the quadratic equation $x^2 - 11x + 22 = 0$ are $x = 3$ or $x = 6$. What is the base of the numbers?	
Q2	(20 marks)	
(a)	The voltage across silicon diode at room temperature is 0.61 V when 2.5 mA current flows through it. If the voltage increases to 0.7 V, then calculate the new diode current.	CO2
(b)	A sample of Ge is made p-type by adding acceptor atoms at a rate of one atom per $5 \times 10^8/\text{m}^3$. If density of Ge atoms is $2.4 \times 10^{25}/\text{m}^3$, determine electron and hole concentration in the doped semiconductor, $n_i = 2.5 \times 10^{15}/\text{m}^3$.	
(c)	Explain the forward and reverse bias condition of a PN junction diode with its V-I characteristics.	
Q3	(20 marks)	
(a)	A crystal diode having internal resistance of 25Ω is used for half wave rectification. If a 240 V, 50Hz is applied to the primary winding of the transformer, used in the rectifier, with turns ratio of 10:1 and load resistance is 1 k Ω , then find: (i) I_m, I_{dc}, I_{rms} (ii) ac power input (iii) dc power output (iv) dc output voltage (v) Ripple factor (γ)	CO3
(b)	Derive an expression for ripple factor (γ) of full wave rectifier.	
(c)	Explain the working of half wave rectifier with capacitor filter and draw appropriate circuit diagram as well as input/output waveforms. Also, derive the RMS current for half wave rectifier without filter.	
Q4	(20 marks)	

(a)	With the appropriate circuit diagram, explain the current amplification factor, total output current, input and output characteristics of Common Emitter Configuration of BJT.	CO2
(b)	What is Bipolar Junction Transistor? What are its different modes of operations.	
(c)	(i) Derive the relationship between I_{CEO} and I_{CBO} . (ii) Calculate α and β of a transistor for which I_C is 5 mA, I_B is 65 μ A and I_{CBO} is 1.7 μ A.	
Q5	(20 marks)	CO3/CO4
(a)	Explain inverting and non-inverting amplifier in detail.	
(b)	Derive an expression for output voltage of difference amplifier.	
(c)	Derive an expression for output voltage of summing amplifier.	